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Venture Vision: Data Center to AI Factory

While startups from clean tech to semiconductors are benefitting from AI capex, we see interesting business models emerging that increasingly view data centers as integrated AI factory systems that require coordination of everything from power to data movement orchestration.

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Key Takeaways

- We map into four 'pathways' US startups innovating power, compute or their integration that have together raised over $50bn since the start of 2024.
- Increasingly-integrated companies remain fewer by count, but are beginning to achieve unicorn valuations and are some of the faster risers.
- Seed financing and investor backing show an expansion from compute into power in 2026.
- Capital is flowing to three very different time-to-scale horizons from near-term execution capacity to novel compute siting models.
- Power is the key constraint, but localized environmental, social and policy concerns can increase the value of sustainable and/or efficient solutions.

Tech Diffusion
A Morgan Stanley Research Key Theme of 2026

One of our predictions for the year was the convergence of power and AI, and this direction is becoming clearer in the AI Infrastructure innovation economy. As AI capex supports technologies from clean energy to semiconductors, new business models are emerging around the coordination of increasingly complex infrastructure. AI neoclouds offer an early example: many gained rapid market validation by pairing access to constrained resources such as power and compute with the ability to deploy capacity quickly. **Increasingly, however, the opportunity is moving beyond scarcity alone toward integration: coordinating power, compute, networking, cooling, and data movement as part of a broader AI factory system.**

We see a broadening opportunity set for startups that address various bottlenecks and/or attempt to future-proof AI factories in the following pathways:

(1) Accelerating access to power: *Generation Buildout (nuclear, fusion, geothermal, onsite generation and powered campuses), Firming & Storage, Delivery & Interconnection, and Distributed Energy & VPPs.*

(2) Increasing useful compute delivered per unit of power: *Compute Silicon,*

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Interconnect & Fabric (photonics, networking), Chip & Rack Thermal, Memory & Storage, Power Delivery & Rack Systems, and GPU Capacity Platforms.

(3) Re-designing power and compute as a single integrated system: *Grid-Interactive Compute Orchestration and Co-Located Grid Integration.*

(4) Reimagining the data center itself through new physical architectures and locations: *Distributed & Modular Data Centers, Environment-Coupled Facilities, Space-Based Data Centers (orbital compute and storage), and Energy-Integrated Compute Platforms.*

Venture funding for this universe has accelerated sharply in 2026, reaching $33.5 billion by early October, nearly double the 2025 total. Most of it has followed the two scarcest inputs to AI infrastructure, compute performance per watt and firm power: Pathways 1 and 2 together account for more than 80% of 2026 capital. Concentration is greatest in GPU capacity and energy-integrated compute, where companies including FluidStack, Groq, GMI Cloud and TensorWave, received about one-third of 2026 capital. **The fastest risers include AI-tailored chips and interconnect in Pathway 2, on-site power in Pathway 1, and integrated power-and-compute models in Pathways 3 and 4**.

Both seed financing and investor activity show an expansion from compute into power. Seed activity has moved from compute toward power and the bottlenecks surrounding it: subsea geothermal (Endurance Energy), on-site power (Transformative American Resources), transformers (Fluxco), magnetics (CorePower Magnetics) and advanced nuclear (Applied Atomics) raised the largest seed rounds in our universe this year. Investor participation shows the same expansion. In 2026, 73% of power and compute rounds included an investor already backing the other pathway, up from 49% in 2024 and 16% before 2021. Most of these moves ran from compute into power: about three times as many investors entered power from compute in 2025–26 as the reverse.

Capital is flowing to three very different time-to-scale horizons at once. Near-term execution capacity is being funded through GPU-native cloud providers, which are deploying compute today. Longer-dated energy solutions are being funded through fusion and advanced nuclear, which raised about $4.5 billion in 2026 even though commercial output remains years away. And entirely new siting models for the data center are being funded through orbital and modular facility developers such as Starcloud, Cowboy Space and Armada, which together raised close to $1 billion in 2026.

The next phase of validation should increasingly shift from capital raised to capacity deployed. Commercial agreements, regulatory approvals, first deployments and evidence of competitive unit economics will matter more than financing alone. Execution risk will remain technology-specific: nuclear depends heavily on licensing, fuel and construction execution, while grid and power technologies face interconnection, utility qualification and long procurement cycles. **At the same time, U.S. data-center growth is becoming increasingly localized.** Cushman & Wakefield notes that development outcomes are now being shaped more heavily by local power-allocation frameworks, transmission constraints, infrastructure requirements and regulation, even as underlying demand remains strong. Energy

policy is becoming part of that divergence and states are treating the technologies in our universe differently. **As a result, deployment economics, and the relative attractiveness of different power, cooling and grid technologies, are likely to become increasingly market-specific.**

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The valuation information provided here is included for illustrative purposes only. The information is based on publicly available information (funding rounds, company disclosure, third-party vendors, etc.) and was not produced or endorsed by Morgan Stanley Research.

Venturing Around Bottlenecks

One of our predictions for the year was the convergence of power and AI, and this direction is becoming clearer in the AI Infrastructure innovation economy. As AI capex supports technologies from clean energy to semiconductors, new business models are emerging around the coordination of increasingly complex infrastructure. AI neoclouds offer an early example: many gained rapid market validation by pairing access to constrained resources such as power and compute with the ability to deploy capacity quickly. **Increasingly, however, the opportunity is moving beyond scarcity alone toward integration: coordinating power, compute, networking, cooling, and data movement as part of a broader AI factory system.**

Rising compute demand, evolving AI architectures and infrastructure constraints are driving the need for solutions purpose-built for AI data centers. While new computing hardware continues to improve performance and efficiency, rack-scale systems are becoming more power-dense and infrastructure-intensive, increasing requirements across memory, networking, cooling and power delivery. As workloads shift toward inference, interactive and agentic use cases, performance increasingly depends on the full system rather than the accelerator alone. We believe the key constraint remains the power and cooling operators can deliver per rack (see Racking Up the Power Shortfall). At the same time, environmental, social and political considerations are increasing the value of solutions that improve efficiency, sustainability and host-community outcomes.

Together, these dynamics are broadening the opportunity set for emerging technologies that address critical infrastructure bottlenecks while future-proofing increasingly integrated AI factories across four pathways: **(1)** accelerating access to power, **(2)** increasing useful compute delivered per unit of power, **(3)** re-designing power and compute as a single integrated system, and **(4)** reimagining the data center itself through new physical architectures and locations.

(1) Accelerating access to power: *Generation Buildout (nuclear, fusion, geothermal, onsite generation and powered campuses), Firming & Storage, Delivery & Interconnection, and Distributed Energy & VPPs.*

(2) Increasing useful compute delivered per unit of power: *Compute Silicon, Interconnect & Fabric (photonics, networking), Chip & Rack Thermal, Memory & Storage, Power Delivery & Rack Systems, and GPU Capacity Platforms.*

(3) Re-designing power and compute as a single integrated system: *Grid-Interactive Compute Orchestration and Co-Located Grid Integration.*

(4) Reimagining the data center itself through new physical architectures and locations: *Distributed & Modular Data Centers, Environment-Coupled Facilities, Space-Based Data Centers (orbital compute and storage), and Energy-Integrated Compute Platforms.*

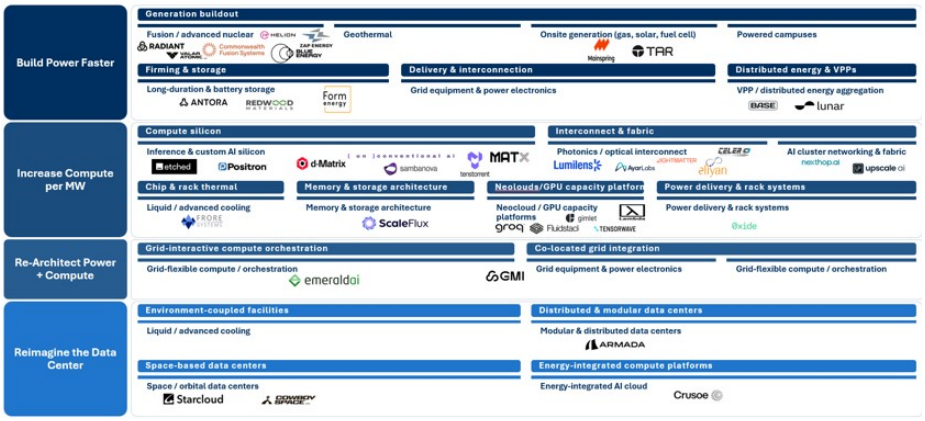
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While it is important to note that some AI infrastructure startup technologies were pre-existing and are now benefiting from AI capex (such as battery storage, advanced nuclear and geothermal), some are becoming increasingly AI-driven, such as networking, liquid

cooling and power quality. Pathways 3 and 4 highlight highly integrated solutions arising largely for AI data centers, treating the grid, onsite generation, storage and the data centre as one co-designed system rather than separate problems.

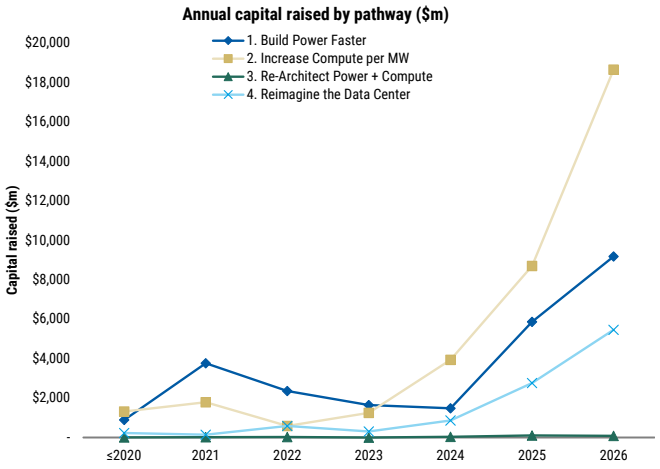
We map currently private US-headquartered VC-backed companies addressing these pathways across specific technology sub-layers and highlight those with valuations of $1 billion or more. **Custom silicon and optical interconnect stand out by number of unicorns, while energy-integrated compute contains the highest-valued company in the current universe.**

Exhibit 1: Unicorns Among VC-Backed AI Infrastructure Innovation Pathways



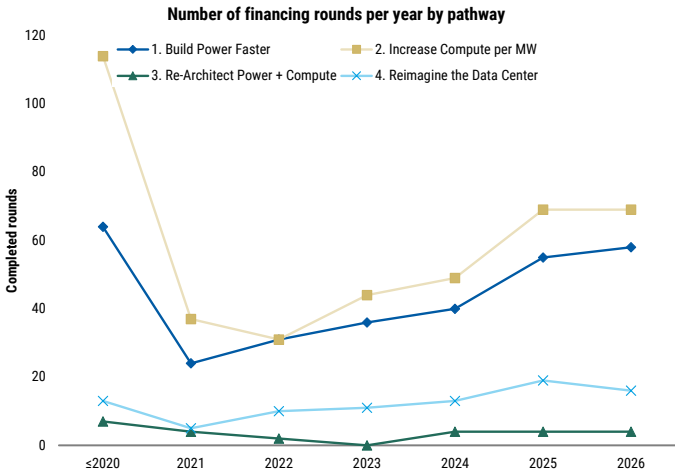
Source: Pitchbook. Morgan Stanley Research. Based on Latest Known Post-Valuation.

Exhibit 2: Annual Capital Raised: By Pathway



Source: PitchBook, Morgan Stanley Research. Completed financings.

Exhibit 3: Financing Rounds: By Pathway



Source: PitchBook, Morgan Stanley Research. Completed financings.

Since the start of 2024, every exit recorded in the universe has come from Pathways 1 and 2, reflecting new firm power generation and energy-efficient, high-bandwidth, thermally managed AI compute systems. Power exits were public listings, including:

- X-energy, which develops small modular reactors with its own TRISO fuel.
- Fervo, which develops enhanced geothermal using drilling techniques from shale.
- Oklo, which is building compact fast-neutron fission reactors and listed through a SPAC.

- Deep Fission, which places small reactors deep in boreholes.

Whereas most compute exits have been acquisitions:

- Marvell bought Celestial AI (photonic interconnect) and XConnTech (CXL and UALink switching) in February 2026.
- AMD bought Enosemi (co-packaged optics).
- Flex bought Jetcool (direct-to-chip liquid cooling).
- Trane bought LiquidStack (immersion cooling).
- SoftBank bought Ampere (energy-efficient server CPUs).

Three compute companies went public instead: Cerebras (wafer-scale AI chips), CoreWeave (GPU cloud) and Astera Labs (connectivity semiconductors). The IPO candidates each sell a complete product or service directly to customers at scale: full AI systems (Cerebras), rented GPU capacity (CoreWeave), and connectivity chips already shipping in volume (Astera). The acquired companies mostly make enabling components that are worth more inside a larger platform than on their own, such as optics, switching fabric and cooling. Their buyers were semiconductor and industrial incumbents with existing customers and distribution.

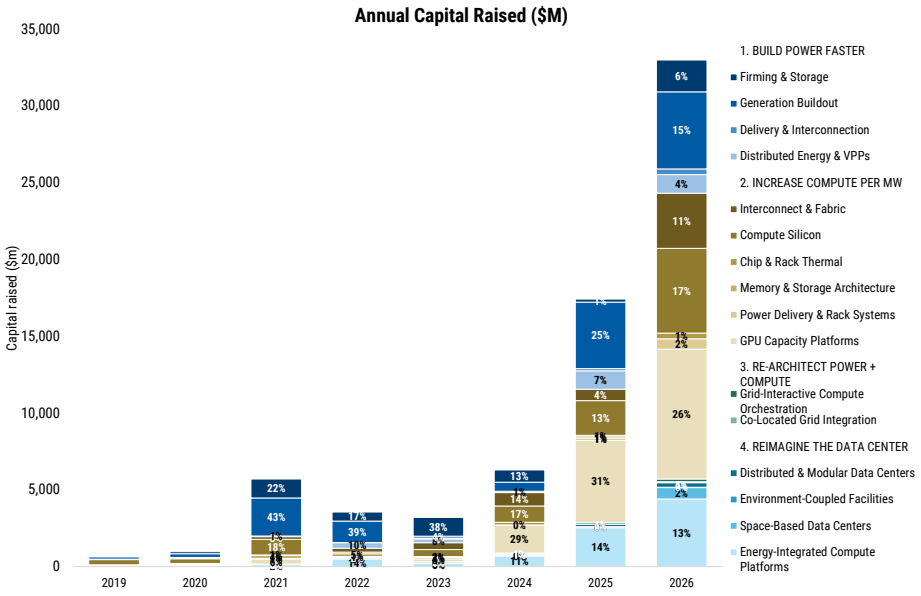
Venture capital has also mostly followed the two scarcest inputs to AI infrastructure: compute performance per watt and firm power (Pathways 1 and 2). Compute (Pathway 2) accounts for more than half of 2026 capital, power (Pathway 1) for just over a quarter, and the reimagined data center (Pathway 4) for about a sixth. Over the full history, three technologies stand out: GPU capacity platforms (neoclouds, about a quarter of cumulative capital), fusion and advanced nuclear, and custom AI silicon (each about one-sixth). The two hardware categories are funded in distinct ways:

- *Silicon:* Capital is concentrated in a small group of unicorn challengers to the GPU (Etched, SambaNova, Tenstorrent, Positron and MatX), which took more than 85% of 2026 silicon capital, while the smaller chip developers raised comparatively little this year.
- *Power generation:* The tier of companies valued below $1bn is also well funded, with $1.3 billion raised in 2026 across 19 rounds. Geothermal, which has no unicorn among its five companies, accounts for nearly half of that tier.
- Breadth beyond the leaders: In 2026, capital extended into the supporting layers. Networking (Nexthop AI, Upscale AI), optical interconnect (Lumilens, Ayar Labs, Celero and Eliyan), long-duration storage (Form Energy and Antora Energy) and geothermal each raised several times their 2025 totals. Networking and optical interconnect are notable for the number of companies financed and the absence of a single dominant recipient.

Participation remains wide, with more than half of the universe raising capital in 2026. However, the unicorns received 86% of the year's capital. Capital has concentrated in capacity platforms. **Concentration is most pronounced in GPU capacity and energy-integrated compute**, where companies (including FluidStack, Groq, GMI Cloud and TensorWave) received about 35% of 2026 capital. **The largest of these businesses are financed in the manner of infrastructure, through large equity rounds supplemented by debt and capital expenditure financing** (Exhibit 6). This distinguishes them from the

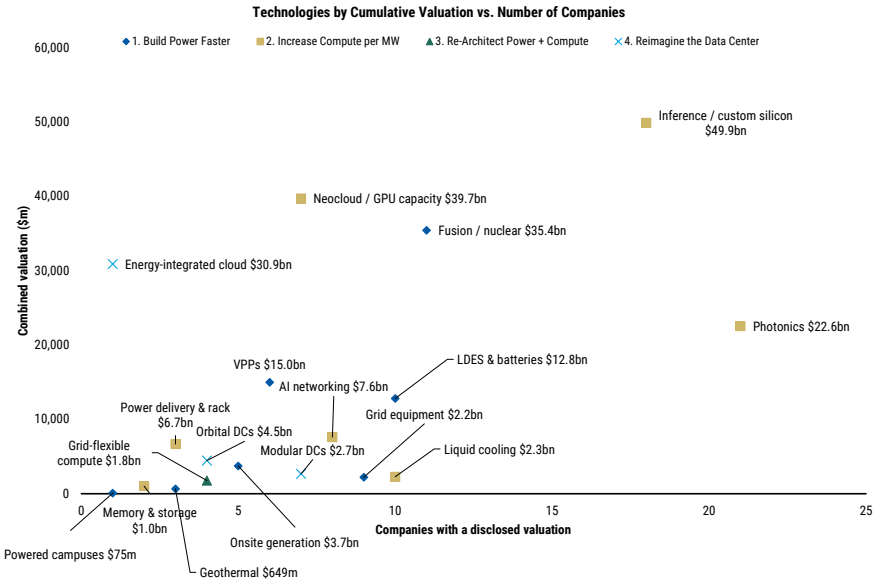
chip and photonics developers, which rely primarily on venture equity.

Exhibit 4: Annual Capital Raised by Sub-Layer



Source: Pitchbook, Morgan Stanley Research. Completed financings.

Exhibit 5: Cumulative Valuation vs Concentration by Technology



Source: Pitchbook, Morgan Stanley Research. Latest known valuation (post-money), associated with completed rounds.

Highly integrated approaches are fewer by number of companies, but break-out businesses have reached unicorn status, validating investor appetite for models that combine power and compute in a single system:

- **Pathway 3**: Can compute load itself respond to the power system?
  - *Grid-interactive compute orchestration.* Emerald AI, which modulates compute demand in response to grid conditions, reached a valuation of about $1 billion. **Its thesis is that certain AI workloads can become flexible loads, allowing data centers to function partly as grid assets rather than only as power**

consumers. Phaidra (~$670 million) applies the same logic inside the facility, using AI agents to co-optimize power, cooling and workload. Verrus takes it a step further, building data centers designed from the outset to flex their power use.

- *Co-located grid integration.* GridCARE (~$334 million) and Camus Energy (~ $50 million) help data centers **connect faster by using spare grid capacity that exists outside peak hours**. Because that only works if the data center agrees to reduce its load when the grid is stretched, both sit in Pathway 3, where compute responds to the power system, rather than Pathway 1, where new supply is added for a load that stays fixed.

- **Pathway 4:** Bring compute to the power, instead of power to compute.
  - *Energy-integrated compute platforms.* This segment covers companies that own both the power and the compute: they secure and integrate their own energy supply, build and operate the data center campus, and sell GPU capacity on top. Because the model spans GPU capacity (Pathway 2) and new approaches to where and how facilities are built (Pathway 4), our precedence rule places it in Pathway 4. That end-to-end approach separates these platforms from neoclouds that run GPU clusters in third-party data centers on power supplied by partners or landlords. **By bringing power and compute online together, energy-integrated platforms address an immediate bottleneck on a relatively short deployment timeline, and their valuations reflect a premium for where and how their data centers are built, not only what goes inside them.** Owning more of the stack gives them more control over execution, but also leaves them more exposed to project delays and the capital demands of building power and facilities alongside compute.
  - *Distributed, modular and orbital facilities.* Starcloud, Cowboy Space, Armada, Rune and Flexnode are **building data centers around alternative form factors and siting, to reduce dependence on conventional constraints on power, land and grid connection**. Starcloud and Cowboy Space are taking compute into orbit, where solar power is near-constant and heat is shed by radiation rather than by water or air cooling.Armada, Rune and Flexnode are deploying prefabricated modular units, several of them liquid-cooled, that can be placed at remote sites, at wind and solar farms (Rune's behind-the-meter model) or wherever power and connectivity already exist. A smaller group couples the facility to its environment rather than its power source: Nautilus Data Technologies, NetworkOcean and Atomarine use ocean or subsea water for cooling, and AVNOS converts data-center waste heat into cooling, water and carbon removal.

Other cross-cutting themes, such as specialized edge AI, defense and community-positive technologies merit monitoring, though their funding share remains small.

- **Edge AI**. Eleven companies span the edge stack, from efficient inference chips (EnCharge AI, Mythic, MemryX, Efficient Computer) to deployed edge facilities (Armada, Greensparc, Northstar). Edge raised $0.4 billion in 2026, a little over 1% of the total. That suggests investors are still backing centralized capacity, even as inference is expected to move closer to users.
- **Defense and infrastructure security**. National-security interest shows up in two

ways. A handful of companies build for defense and mission-critical use: rugged edge data centers (Armada), modular secure facilities (Northstar), portable microreactors to replace diesel at military bases (Radiant) and space-based compute (Starcloud, Sophia Space). These raised $0.8 billion in 2026. More significant is the capital: defense- and intelligence-linked investors appear on the cap tables of companies at every layer of the stack, from GPU clouds and optical interconnect (Ayar Labs) to nuclear (Valar Atomics, backed by Lockheed Martin Ventures) and grid-flexible compute (Emerald AI). In-Q-Tel alone is an investor in 11 companies that together raised $5.5 billion this year. This suggests national-security investors view AI infrastructure as strategic capacity rather than a niche defense market.

- **Community and host benefit.** In addition to general compute efficiency gains, a small group addresses the local cost of data centers through heat reuse and water production (AVNOS, Nautilus, ZutaCore, Karman) and through reuse of existing energy infrastructure (Quaise Energy). The eight companies we tag here have raised about $950 million in total, and $330 million in 2026, about 1% of the year's capital.

Funding the Pathways

2025 breakout rounds. 2025 was the year in which very large rounds became established practice, with a small group of companies raising amounts well beyond the prior norm.

- *Fusion and fission:* Pacific Fusion's $900m Series A, Commonwealth Fusion Systems' $863m Series B2, TerraPower's $650m round and Helion's $425m Series F. Radiant, which develops portable microreactors, also reached unicorn status.
- *Compute*: Groq raised across two rounds ($750m and $1.5bn; now part of the GPU capacity platform layer following its Nvidia licensing agreement), Etched completed its first very large financing ($500m) and d-Matrix closed a $275m Series C. The most notable was Unconventional AI's $475m seed round, an exceptionally large initial financing for a company founded in 2025 and now valued at $4.5bn.
- *Distributed power:* Base Power's $1.0bn Series C made it the largest raise among the distributed-energy companies.

Novel form factors: Starcloud and Cowboy Space (orbital compute) and Armada (modular edge data centers) are small in absolute dollars but are each valued above $2bn, indicating that investors are paying for optionality on where compute capacity will ultimately be located.

Fastest risers: Eight of the 40 unicorns were founded in 2024 or later. They span every pathway, but most are in compute:

- (1) Build Power Faster: Transformative American Resources, which builds on-site renewable power systems for data centers. It was founded in 2026 and was valued at $1.0bn by September 2026, the fastest rise in the universe.
- (2) Increase Compute per MW:
  - Unconventional AI, building a new energy-efficient AI chip architecture: founded 2025, about $4.5bn by December 2025.

    - Upscale AI, AI networking: founded 2025, $2.0bn in June 2026.
    - Lumilens, photonic interconnect: founded 2024, $5.5bn.
    - Nexthop AI, AI cluster networking: founded 2024, $4.2bn.
    - Celero, optical signal processing: founded 2024, $3.1bn.
- (3) Re-Architect Power + Compute: Emerald AI, which makes data-center load-flexibility software. It was founded in 2024 and reached $1.05bn in August 2026.
- (4) Reimagine the Data Center: Starcloud, which builds orbital data centers. It was founded in 2024 and reached $2.3bn in August 2026.

Seed rounds. Typical seed rounds remain modest, in the mid-single-digit millions. The exception is a small number of very large seeds in 2025, led by Unconventional AI ($475m) and Upscale AI ($100m), both in compute and both now unicorns. These accounted for about three-quarters of that year's seed capital and have not been repeated at the same scale in 2026.

- Unconventional AI develops a new chip architecture for AI that moves away from the traditional design of separate processor and memory, aiming for much greater energy efficiency.
- Upscale AI builds open-standard, high-performance networking that connects AI computing clusters.

This year, seed activity has moved from compute toward power and the bottlenecks surrounding it, including subsea geothermal (Endurance Energy), on-site power (Transformative American Resources), transformers (Fluxco), magnetics (CorePower Magnetics) and advanced nuclear (Applied Atomics). In Pathway 3, Emerald AI (load-flexibility software for data centers) raised a $25m Strategic Expansion Round in March 2026. Energy Impact Partners led the round, and strategic investors including NVentures, GE Vernova, Siemens, Eaton and Samsung joined. Emerald followed it with a $150m Series A in August at a $1.05bn valuation.

Most frequent financings included Ayar Labs (optical I/O), Antora Energy (thermal batteries) and Form Energy (iron-air storage) have each completed well over a dozen financings. This reflects the high capital requirements of long-duration hardware developers as well as investor willingness to support scaled leaders.

Investor backing across the universe is broad but split by layer. The most frequent backers include In-Q-Tel, the U.S. Department of Energy, Valor Equity Partners, Climate Capital, Andreessen Horowitz, Y Combinator and Nvidia, and only about half of the most active names are traditional VCs. The rest are government, strategic, accelerator and crossover capital.

- Compute efficiency (Pathway 2) has the deepest bench of repeat investors, with about 85 investors backing three or more of its companies. It is led by semiconductor-focused funds such as Foothill Ventures and Maverick Silicon, alongside Atreides, Nvidia and Arm, and chip and hyperscaler corporates are heavily involved.
- Power generation (Pathway 1) leans on climate and impact capital, mainly Breakthrough Energy, Valor Equity Partners, Climate Capital, Energy Impact Partners and Lowercarbon Capital, and has far fewer repeat backers (about 30).
- Cross-industry interest is building in Pathway 3, with strategic investors such as

Nvidia, GE Vernova and Siemens backing Emerald AI.

- Only the U.S. Department of Energy and In-Q-Tel invest in all four layers.

- Among the $1bn+ companies, Andreessen Horowitz, Atreides, Sequoia, Valor Equity Partners and Alumni Ventures are the most frequent backers.

Investor overlap between power and compute has become the norm. In 2026, 73% of power and compute rounds with disclosed investors, accounting for 89% of their capital, included at least one investor that had already backed a company in the other pathway (Exhibit 7). That is up from 49% in 2024 and 16% before 2021. These investors increasingly lead the round as well as join it: 40% of 2026 rounds with a named lead were led by such an investor, compared with about 5% in 2021–22. **The flow runs mainly from compute investors into power.** In 2025–26, about three times as many investors made their first move from compute into power as made the reverse move, among them SoftBank, Microsoft, NVIDIA, Sequoia and Atreides.

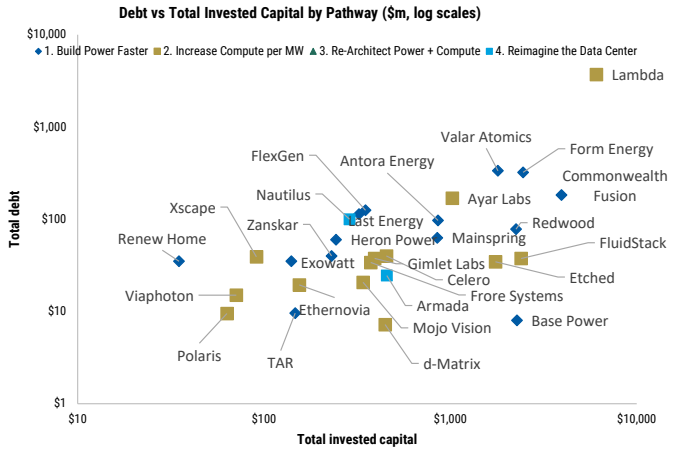
Pathways 3 and 4 are where these investor bases converge: in 2026, 72% of their rounds included an investor that already backed both a power and a compute company, up from 14% before 2023, drawing in compute-side backers such as NVIDIA, Sequoia and Microsoft alongside power-side investors such as Breakthrough Energy and Energy Impact Partners.

Decacorns. Several companies are now valued above $10 billion including Etched, FluidStack, Helion, Base Power, SambaNova and Commonwealth Fusion Systems. Together decacorns account for more than half the combined valuation of the 40 unicorns.

Down rounds are rare in the universe. Of the 101 companies with two priced rounds to compare, 90 raised their latest round at a higher valuation, five were flat and only six were down (about 6%).

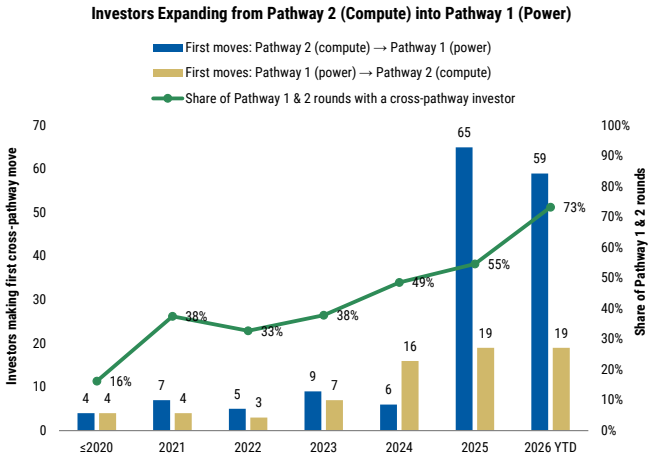
- Form Energy: Its August 2026 $750m Series G was priced at $1.75bn pre-money ($2.5bn post-money), against a $3.3bn post-money valuation for its 2024 Series F, about 47% lower. The round came with a $270m revolving credit facility.
- Smaller resets: NuCube Energy, Camus Energy, Aesir Technologies and Mixed-Signal Devices each repriced 10–33% lower, on rounds of under $70m. Several of these followed 2021-era valuations.

Exhibit 6: Total Debt vs Total Capital Raised



Source: Pitchbook, Morgan Stanley Research

Exhibit 7: Investor Backing & the Pathways



Source: Pitchbook, Morgan Stanley Research

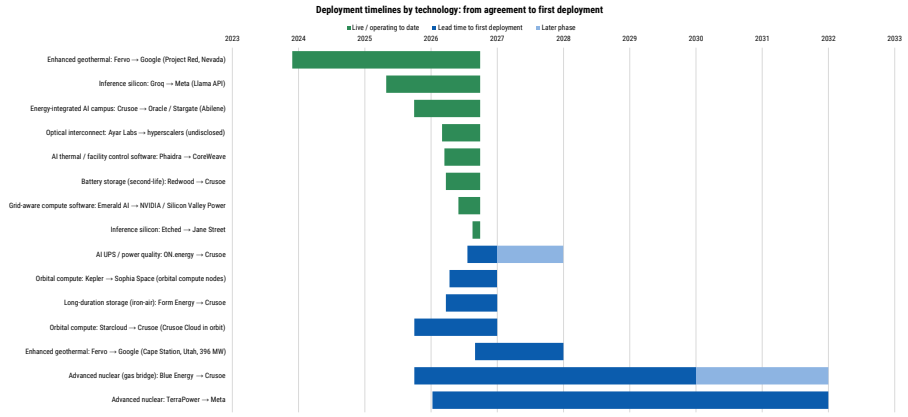
Commercialization & Time-to-Scale

Capital is flowing simultaneously to three very different time horizons. Near-term execution capacity is being funded through GPU-native cloud providers, which are deploying compute today. Longer-dated energy solutions are being funded through fusion and advanced nuclear, where commercial output remains years away. And entirely new siting models for the data center are being funded through orbital and modular facility developers such as Starcloud, Cowboy Space and Armada, which together raised close to $1 billion in 2026.

The key distinction is between commercial validation and commercial scale. AI neoclouds, battery storage, networking and some cooling technologies can enter the current buildout relatively quickly. Nuclear and fusion developers, by contrast, may secure hyperscaler partnerships years before they have meaningful operating capacity. Orbital data-center startups are earlier still on the commercialization curve while water-integrated facilities are in exploratory stages in the US.

We highlight AI-related commercialization evidence vs deployment timelines for companies in our universe:

Exhibit 8: Commercial Agreement Deployment Timelines



Source: Morgan Stanley Research. Company Announcements.

Commercial evidence and time-to-scale outlooks by pathway:

(1) Build Power Faster. Offtake and capacity commitments demonstrate commercial demand: Google's 200 MW offtake from Commonwealth Fusion's first ARC plant and 396MW PPA with Fervo Energy's geothermal systems, Meta's agreement supporting up to eight TerraPower Natrium plants, and Form Energy's iron-air storage reservation. Sunrun recently announced it will supply capacity from thousands of residential storage-plus-solar systems for Voltus' Bring Your Own Capacity (aggregating distributed energy) program for AI hyperscalers.

Nuclear outlook: Our teams recognize that nuclear provides scalable, carbon-free around-the-clock baseload power needed by AI data centers over the 2030s. However, we exclude nuclear from solving near-term (through 2028) power shortfalls due to regulatory, permitting, and multi-year construction lead times. Instead, our Clean Tech & Utilities team prefers monitoring near-term derisking milestones for advanced nuclear developers like X-energy, specifically supply-chain agreements (e.g., fuel fabrication and high-assay low-enriched uranium [HALEU] sourcing), expanded Department of Energy Advanced Reactor Demonstration Program (ARDP) cost-share commitments, and utility/hyperscaler anchor offtake contracts. Operational nuclear power plant co-location (e.g., behind-the-meter arrangements with existing generation) provides only 1 to 3 GW of near-term bridge capacity through 2028.

A recent example of a regulatory milestone includes Blue Energy's September filing of a construction permit application with the NRC for its gas-to-nuclear Texas campus. Earlier this year, the NRC granted a construction permit for Terrapower's Natrium plants (first ever for a commercial non-light-water power reactor).

Underappreciated Bridge Technologies (Fuel Cells, Behind-the-Meter Turbines, Storage & CCS): With a projected gross US data center power shortfall of 57 GW (and a net deficit of 33 GW through 2028 after initial grid allocations and builds), hyperscalers and powered shell providers are forced to bypass multi-year utility queue delays via behind-the-meter (BTM) solutions:

- *Solid Oxide Fuel Cells*: Morgan Stanley maintains an OW rating on Bloom Energy (BE). Fuel cells provide rapid 50-to-100-day time-to-power energization on natural gas or hydrogen blends, delivering reliable baseload power while avoiding grid

transmission bottlenecks.

- *BTM Gas Turbines & Reciprocating Engines*: BTM generation equipment (GE Vernova, Cummins, Innio) and power-as-a-service operators (SEI) bridge 11 to 19 GW of near-term demand. Hyperscalers accept higher interim fuel costs because accelerating energization by just one year creates significant economic value.
- *Long-Duration Energy Storage (LDES) & Sodium-Ion:* Multi-day storage technologies (such as Form Energy's iron-air battery chemistry) and emerging sodium-ion chemistries address renewable intermittency and peak shaving without relying on scarce lithium-supply chains, unlocking 24/7 carbon-free capacity matching for hyperscalers by the late 2020s.
- *Carbon Capture & Storage (CCS):* Point-source carbon capture on distributed natural gas plants allows data center operators to deploy high-uptime thermal generation today while complying with evolving regional emissions mandates (such as Virginia's Clean Economy Act and regional greenhouse gas frameworks)

Our autos, utilities, energy and industrials teams believe behind-the-meter power is an architectural shift in how power is consumed beyond large AI training facilities. They expect BTM adoption to broaden as more inference AI data centers get built, physical AI proliferates across hardware. Commercial & Industrial customers facing grid access, reliability and time-to-power constraints will have greater adoption potential as the current investment flywheel, with secondary support from US onshoring, scales, lowers costs and improves reliability of emerging technologies (fuel cells, packaged microgrids, SMRs in the 2030s).

The teams outline lower costs vs time-to-power vs emissions and other factors as a framework to assess which technologies are the most attractive solutions in the near-term. They determine that Bloom Energy fuel cells are the most attractive BTM power source today based on the composite scale.

They expect aggregating existing distributed energy to become increasingly relevant to data centers, and see convergence of distributed energy and compute.

Powering AI: Racking Up the Power Shortfall (21 Sep 2026), Autos & Shared Mobility / Utilities & Clean Tech / Energy / Industrials: Living on the Edge: The Rise of Distributed Power (6 Oct 2026)

(2) Increase Compute per MW. Commercial validation in this layer is established by shipped production hardware and named anchor customers, exemplified by Etched shipping its first specialized Sohu transformer-ASIC rack to Jane Street. Exits are dominated by strategic M&A: of the ten exits in this pathway, seven were acquisitions, six of them by public strategic buyers such as Marvell, AMD and SoftBank, which acquired interconnect, photonics and compute-silicon IP alongside cooling assets bought by industrial firms such as Flex and Trane. **Our teams view low-latency inference, architectural disaggregation, and specialized silicon and interconnects as critical enablers for overcoming memory and interconnect constraints**.

Our semiconductor team expects memory shortages to constrain AI builds for the rest of this cycle. AI will absorb essentially all available supply, so the industry is looking for ways to become more memory-efficient rather than waiting for new DRAM capacity. The team sees four main workarounds.

- *Selective de-speccing*: lowering HBM stack heights, as in NVIDIA's Rubin move from 12-high to 8-high, and reducing rack-level LPDDR. This lets compute vendors meet volumes but shifts pressure to NAND and to larger scale-up domains.
- *Disaggregated inference*: pairing compute-dense GPUs or Trainium for prefill with memory-bandwidth-optimised chips for decode. Cerebras (with AMD in 4Q26 and with AWS on Bedrock in 1Q27) and NVIDIA (through Groq) are the clearest beneficiaries, and d-Matrix and SambaNova are building similar heterogeneous systems.
- *On-chip SRAM*: historically far more expensive than DRAM, but no longer at today's DRAM prices.
- *CXL memory pooling*: letting multiple processors share a common memory pool. Astera Labs' Leo controllers are expected to ramp at two US hyperscalers in 2027, and Marvell sees CXL as a business of more than $1 billion by around 2028.

Semiconductors: How can the AI ecosystem work around memory bottlenecks? (5 Oct 2026), Cerebras Systems: Cerebras posts another strong quarter (13 Aug 2026), Cerebras Systems: Supernova Recap: Cerebras Unveils CS 4 (19 Aug 2026), Astera Labs Inc: Substantial upside driven by the Scorpio X ramp (5 Aug 2026)

Optical Potential & Scale-Up Interconnects: Our networking and semiconductor teams emphasize that interconnect complexity and electrical I/O power dissipation at copper boundaries make optical connectivity inevitable as AI clusters expand beyond single-rack scale-up domains. However, their time-to-scale analysi, further reinforced by ECOC 2026 takeaways, indicates a phased, multi-stage transition:

- Near-Term (2026-2027): *Copper Inside Racks, Light Across Buildings*
  - Pluggable optical transceivers (scaling to 800G and 1.6T) and active copper cables (supported by Astera's Aries retimers and Scorpio switches) dominate rack-level scale-up fabrics. Optical revenue begins scaling through fiber-attached transceivers and Near-Package Optics (NPO).
- Medium-to-Long-Term (2028+): *Bringing Optics Directly to the Chip*
  - As hyperscalers migrate toward Co-Packaged Optics (CPO) and optical circuit switching (OCS) through initiatives such as the Open CPX Multi-Source Agreement, optics will directly address latency, thermal density, and interconnect power limits. This transition expands the scale-up networking TAM to $73 billion, dramatically lifting token throughput per megawatt.

Technology: Bringing More Light to Scale-Up Networks: An Updated Scale-Up Primer (13 Jul 2026), Optical: ECOC First Take: Full Steam Ahead (23 Sep 2026)

(3) Re-Architect Power + Compute. Commercial evidence is operating deployments inside live AI infrastructure: Emerald AI's first commercial multi-megawatt NVIDIA DSX Flex deployment with Silicon Valley Power, Phaidra scaling its AI thermal-management agents across CoreWeave's fleet.

We previously outlined how the pure power bottleneck expanding to power flexibility, volatility management, and time-to-power. Unlike flat baseload consumption typical of long training runs, scaling AI inference and multi-tenant workloads introduces severe power draw volatility and unpredictable demand spikes. Emerald AI's multi-megawatt NVIDIA DSX Flex deployment with Silicon Valley Power operationalizes this exact

principle: compute load is throttled or rescheduled in real time based on utility grid strain, demonstrating that AI clusters can behave as flexible grid assets and secure faster interconnection approvals without waiting for traditional transmission upgrades.

Power for AI: Flexible Power – The Next Wave of Growth in AI (17 Mar 2026)

(4) Reimagine the Data Center. Commercial evidence in this layer is uneven across technology types. It is strongest where reimagined facilities connect to conventional data-center delivery and thinnest in orbital and water-integrated architectures.

- *Energy-integrated and modular facilities.*
  - Armada builds containerized modular data centers for edge and remote deployment. Fossefall has placed an order for the first of five Leviathan modular data centers for a plan to build more than 1 gigawatt of Nordic-owned AI capacity by 2030.
  - ECL provides off-grid, hydrogen-powered modular data centers as a service. It has operated a hydrogen-powered AI facility in Mountain View, California since 2024. In July 2026 it announced a partnership of more than 300 MW with PowerCell, a fuel-cell manufacturer, which disclosed a related firm order.
- *Orbital compute and storage*. Starcloud, which designs data centers in space powered by continuous solar energy, has a deployment agreement and an in-orbit demonstration. Other orbital data-center developers remain largely pre-commercial, although several have begun demonstration mission.
  - Cowboy Space recently launched its first satellite, and maintains production targets for 2028. Sophia Space, which develops modular, thermally managed compute hardware for orbit, is scheduled to deploy edge compute nodes on Kepler Communications satellites in late 2026.
  - Nvidia has partnerships with Starcloud, Sophia Space and others to deploy their compute in space.
- *Water-integrated and other facilities*. Subsea and offshore data-center concepts remain largely in the pilot, proposal, and feasibility-study stages in the U.S.

Our teams have profiled Starcloud here: Starcloud: Data Centers in Space - AI Goes Orbital (27 Jan 2026)

Orbital Compute Outlook: Our space and technology teams emphasize that while orbital compute circumvents terrestrial permitting bottlenecks, water cooling constraints, and power grid interconnection queues, **near-to-medium-term commercial adoption will center on orbital edge AI** (processing Earth observation imagery and sensor data in orbit to reduce downlink bandwidth and latency) rather than replacing terrestrial hyperscale training clusters. SpaceX (OW) plans to deploy its initial "Starmind" compute satellites by 2028, with the team modelling GW-scale compute by 2030 and deliver cost parity against terrestrial clusters by 2031. For the broader future of AI infrastructure, commercial deployment follows a three-stage evolution:

  - *Three-Stage Roadmap:*
    - Stage 1 (Near-Term): Orbital Edge AI for real-time sensor and imagery processing (e.g., Starcloud-1, Adaspace)
    - Stage 2 (Medium-Term): Distributed Orbital Cloud via inter-satellite

optical mesh networks.
- Stage 3 (2030s+): Scaled space-based training infrastructure enabled by heavy-lift launch cadence.

SpaceX: Stefan-Boltzmann: Das ist super cool (21 Jul 2026), Tech Bytes: Space Technology – Orbital Computing (7 Jul 2026), Space Exploration Technologies Corp.: A 'Heavenly' Hedge to DC NIMBYism? (16 Jul 2026), SpaceX: AI's Final Frontier; Initiate at Overweight, PT $300 (7 Jul 2026),

Monitoring the (Bottleneck) Situation

The next phase of validation should increasingly shift from capital raised to capacity deployed. Commercial agreements, regulatory approvals, first deployments and evidence of competitive unit economics will matter more than financing alone. Execution risk will remain technology-specific: nuclear depends heavily on licensing, fuel and construction execution, while grid and power technologies face interconnection, utility qualification and long procurement cycles.

At the same time, U.S. data-center growth is becoming increasingly localized. Cushman & Wakefield notes that development outcomes are now being shaped more heavily by local power-allocation frameworks, transmission constraints, infrastructure requirements and regulation, even as underlying demand remains strong. Power availability, permitting, water access, transmission capacity and infrastructure readiness therefore vary materially by market, while states and localities are adopting increasingly differentiated approaches to new development. **Energy policy is becoming part of that divergence and states are treating the technologies in our universe differently.** For example, New Jersey has opened the door to new nuclear, and Virginia has turned to heat reuse.

Exhibit 9: US State Policy Themes & AI Factory Pathways

POLICY THEME AND STATE ACTIONS	LAYER 1 Build Power Faster	LAYER 2 Increase Compute per MW	LAYER 3 Re-Architect Power + Compute	LAYER 4 Reimagine the Data Center
1 Advanced nuclear and SMRs **IL** executive order: at least 2 GW of new nuclear, construction by 2033 **NY** pathway to 4 GW, plus at least 1 GW at NYPA **NH** executive order: roadmap to advanced nuclear	Fission developers **Oklo, X-energy, Valar Atomics, Radiant, Aalo, TerraPower, Deep Fission, NuCube, Blue Energy**	–	–	–
2 Heat reuse and water **VA** EO 22 and HB 323: reclaimed water and waste-heat reuse **PA** EO 2026-05: water recycling and closed-loop cooling	–	Liquid cooling **Green Revolution Cooling (GRC), ZutaCore, Accelsius, Frore Systems**	–	Low-water facilities **Nautilus Data Technologies, AVNOS**
3 Backup power and storage **PA** strict limits on diesel and gas backup generators **MD** review of backup-generator emissions and cooling **VA** higher storage mandates; 10+ hour targets	Storage and power quality **Form Energy, Antora Energy, Redwood Materials, FlexGen, Sorbiforce**	–	–	–
4 Self-supply and flexibility **NY** Energize NY: pay more or supply own power **VA** on-site generation; voluntary demand flexibility	On-site power **Transformative American Resources, Mainspring**	–	Load flexibility **Emerald AI, Camus Energy**	Power-integrated facilities **Crusoe, ECL**

Source: Morgan Stanley Research, Local government announcements. Latest known valuation (post-money), associated with completed rounds.

As a result, deployment economics, and the relative attractiveness of different power, cooling and grid technologies, are likely to become increasingly market-specific.

Capital formation is also becoming more specialized around the integrated AI-infrastructure stack. Andreessen Horowitz's $1.1 billion Machine Age Fund targets the physical infrastructure underpinning AI, spanning compute infrastructure and full systems such as data centers and robotics. Elemental Impact's Data Center Innovation Initiative, backed by Amazon, Google, Meta and Microsoft, similarly uses data centers as

deployment environments for emerging energy, electrical, cooling and materials technologies. Together, these initiatives illustrate how AI infrastructure can provide both capital and a route to commercial validation. Federal initiatives are addressing similar bottlenecks.

Federal initiatives in the US are becoming broader in support for infrastructure outside specific verticals, such as nuclear energy. DOE's SPARK (Speed to Power through Accelerated Reconductoring and other Key Advanced Transmission Technology Upgrades) initiative is directing $5.25 billion of federal funding toward transmission upgrades and grid-enhancing technologies that has already committed funding to projects across 26 states. Some of these projects explicitly mention support for AI-enabled Smart grid technologies, flexible power and others. These programs can both alleviate infrastructure constraints and create scaled validation opportunities for emerging grid technologies.

Performance at one layer can create or relieve constraints elsewhere, increasing the value of technologies that improve total-system economics, deployment speed, utilization or reliability rather than optimizing a single component in isolation. **At the same time, this integration raises concentration risk: multiple infrastructure categories are increasingly exposed to the same hyperscaler capex cycle and AI-demand assumptions.** Technologies with diversified customers and applications outside the AI ecosystem should therefore may have a more resilient commercialization profile, but we expect rising demand for compute and broadening AI adoption to support winning AI-tailored solutions in the long-term.

Highly-Valued Companies: 4 Pathways

Exhibit 10: Build Power Faster

Company	Sub-Layer	Valuation ($m)	Valuation date	Lead investors (latest priced round)	Company description
Helion	Generation Buildout	$15,500	Sep 15, 2026	Thrive Capital (Series G, Sep 2026)	Developer of a fusion energy technology designed to generate clean electricity through controlled nuclear fusion.
Base Power	Distributed Energy & VPPs	$13,000	Aug 3, 2026	Valor Equity Partners, Addition, Ribbit Capital (Series D, Aug 2026)	Operator of a residential energy and battery backup service company intended to provide electricity and home power protection.
Commonwealth Fusion Systems	Generation Buildout	$10,000	Aug 19, 2026	No lead named: Plynth Energy, Greycroft, Deep Checks ($1bn round, Aug 2026)	Developer of an energy fusion machine intended to accelerate the path to commercial fusion energy.
Redwood Materials	Firming & Storage	$6,000	Jan 28, 2026	Eclipse Capital (Series E, Jan 2026)	Operator of a battery-materials platform intended to build a domestic closed-loop supply chain for lithium-ion batteries.
Valar Atomics	Generation Buildout	$6,000	Aug 3, 2026	Sequoia Capital (Series B, Aug 2026)	Developer of modular nuclear reactor technology designed to provide scalable clean energy using atomic energy for industrial applications and fuel production.
Antora Energy	Firming & Storage	$2,700	Aug 4, 2026	G2 Venture Partners, Eclipse Capital, Copenhagen Infrastructure Partners (Series C, Aug 2026)	Developer of modular thermal batteries designed to power always-on industrial operations with low-cost renewable energy.
Form Energy	Firming & Storage	$2,500	Aug 12, 2026	T. Rowe Price (Series G, Aug 2026)	Manufacturer of long-duration energy storage systems designed for the electric grid.
Mainspring Energy	Generation Buildout	$1,908	Apr 14, 2025	General Catalyst (Series F, Apr 2025)	Manufacturer of linear power generators intended to bring affordable, reliable, and net-zero carbon grids of the future.
Radiant	Generation Buildout	$1,880	Dec 17, 2025	Global Venture Management (development capital, Jun 2026; no lead named)	Manufacturer of portable nuclear microreactor technology designed to offer a climate-friendly alternative to diesel generators.
Blue Energy	Generation Buildout	$1,580	Jul 16, 2026	VXI Capital Management (Series B, Jul 2026)	Operator of a nuclear fission energy company intended to focus on nuclear power that delivers clean, reliable and affordable power.
Zap Energy	Generation Buildout	$1,100	Jul 30, 2024	Soros Fund Management (Series D, Jul 2024)	Developer of a compact and scalable fusion technology designed to produce sustainable energy.
Transformative American Resources (TAR)	Generation Buildout	$1,000	Sep 10, 2026	Spark Capital (Series A1, Sep 2026)	Provider of energy infrastructure development services designed to construct and maintain systems supporting sustainable energy across the United States.
Lunar Energy	Distributed Energy & VPPs	$1,000	Feb 4, 2026	Prelude Ventures, B Capital (Series D, Feb 2026)	Manufacturer of home energy storage and management hardware designed for residential power independence.
FlexGen (Energy Storage)	Firming & Storage	$770	Jul 15, 2022	Vitol (Series C, Jul 2022)	Developer of energy storage technology designed to deliver utility-scale battery storage projects through software-enabled management and integrated energy storage systems.
Exowatt	Generation Buildout	$695	Nov 13, 2025	8090 Industries, The Merry Ventures, Felicis, MVP Ventures (Series A, Nov 2025)	Operator of a next-generation renewable energy company intended to make sustainable renewable energy always available.

Source: Pitchbook, Morgan Stanley Research. Up to top 15 companies by latest known valuation (post-money), associated with completed rounds.

Exhibit 11: Increase Compute per Unit of Power

Company	Sub-Layer	Valuation ($m)	Valuation date	Lead investors (latest priced round)	Company description
Etched	Compute Silicon	$21,000	Aug 18, 2026	Jane Street ($700m round, Aug 2026)	Developer of rack-scale artificial intelligence inference hardware designed to deliver high-performance computing for production workloads.
FluidStack	GPU Capacity Platforms	$18,000	Sep 3, 2026	Jane Street ($1.5bn round, Sep 2026)	Developer of a cloud-based graphics processing unit designed to turn various internet-connected devices into a cloud server.
SambaNova Systems	Compute Silicon	$11,000	Jul 8, 2026	General Atlantic (Series F, Jul 2026)	Developer of a chips-to-model computing built for an AI platform designed to power machine learning and data analytics.
Together AI	GPU Capacity Platforms	$8,200	Jul 1, 2026	Prosperity7 Ventures, Aramco Ventures (Series C, Jul 2026)	Operator of a cloud-based artificial intelligence infrastructure platform designed to enable organizations to build, train, and deploy AI models at scale.
Oxide Computer	Power Delivery & Rack Systems	$6,346	Aug 4, 2026	US Innovative Technology Fund (Series C, Feb 2026; Series D investors undisclosed)	Manufacturer of an integrated rack-scale computer system designed for on-premises cloud computing.
Lumilens	Interconnect & Fabric	$5,510	Aug 6, 2026	Meritech, Seligman Ventures, Spark Capital (Series C, Aug 2026)	Developer of photonic interconnects designed for scaling AI compute infrastructure.
Positron	Compute Silicon	$5,000	Sep 10, 2026	Valor Equity Partners, NEA, Atreides, GreatThings, SemiAnalysis, Andra Capital (Series	Developer of purpose-built hardware designed for generative artificial intelligence acceleration.
Ayar Labs	Interconnect & Fabric	$5,000	Sep 10, 2026	Neuberger Berman, Artisan Partners (Series E, Sep 2026)	Developer of optical interconnect products designed for artificial intelligence infrastructure.
Lightmatter	Interconnect & Fabric	$4,500	Oct 16, 2024	No lead named: E1 Ventures, Capitalis Partners (Series D, Dec 2025)	Developer of photonic computing and interconnect hardware designed for high-performance data processing and communication in AI infrastructure.
Unconventional AI	Compute Silicon	$4,500	Dec 5, 2025	Andreessen Horowitz, Lightspeed Venture Partners (Seed, Dec 2025)	Operator of an artificial intelligence-focused hardware startup intended to reinvent computer architecture for efficient and scalable artificial intelligence processing.
Nexthop AI	Interconnect & Fabric	$4,200	Mar 10, 2026	Lightspeed Venture Partners, Andreessen Horowitz (Series B, Mar 2026)	Developer of artificial intelligence infrastructure designed for large-scale cloud operators.
Groq	GPU Capacity Platforms	$3,500	Aug 17, 2026	Disruptive, Infinitum Partners (Series A, Aug 2026)	Developer of AI inference platform designed to accelerate artificial intelligence model execution.
Celero	Interconnect & Fabric	$3,118	Sep 8, 2026	CapitalG, Valor Equity Partners, Atreides (Series C, Sep 2026)	Developer of coherent digital signal processor technology intended for high-speed optical transmission.
Gimlet Labs	GPU Capacity Platforms	$3,000	Sep 4, 2026	Andreessen Horowitz (Series B, Sep 2026)	Developer of a cloud platform for artificial intelligence processing intended to streamline the deployment and management of complex agent-based workloads.
Tenstorrent	Compute Silicon	$2,600	Dec 2, 2024	Samsung Catalyst Fund, Samsung Securities, Samsung Venture Investment ($700m round, Dec 2024; Sep 2026 $1.3bn round investors	Developer of artificial intelligence processor architectures and open-source software platforms designed to accelerate deep learning workloads.

Source: Pitchbook, Morgan Stanley Research. Up to top 15 companies by latest known valuation (post-money), associated with completed rounds.

Exhibit 12: Re-Architect Power + Compute

Company	Sub-Layer	Valuation ($m)	Valuation date	Lead investors (latest priced round)	Company description
Emerald AI	Grid-Interactive Compute Orchestration	$1,050	Aug 25, 2026	DCVC, Energize Capital (Series A, Aug 2026)	Developer of an infrastructure flexibility management platform designed to manage the interface between power grids and data centers.
Phaidra	Grid-Interactive Compute Orchestration	$670	Oct 1, 2025	Collaborative Fund (Series B, Oct 2025)	Developer of artificial intelligence-powered control systems intended to help large industrial facilities learn, adapt, and improve performance.
GridCARE	Co-Located Grid Integration	$334	May 14, 2026	Sutter Hill Ventures, John Doerr (Series A, May 2026)	Developer of an AI-driven grid optimization platform designed to help data centers and utilities meet unprecedented electricity demand by rapidly unlocking existing grid capacity.
Camus Energy	Co-Located Grid Integration	$50	Feb 21, 2024	Congruent Ventures, Wave Capital (Series A1, Feb 2024)	Developer of grid orchestration software intended for flexible interconnection and management of electric load and distributed energy resources.
Digital Power Optimization	Co-Located Grid Integration	$4	Jan 27, 2021	New York Digital Investment Group (Series A, Feb 2022)	Provider of energy optimization services intended to create power-dense computing offtake for otherwise stranded energy.

Source: Pitchbook, Morgan Stanley Research. Up to top 15 companies by latest known valuation (post-money), associated with completed rounds.

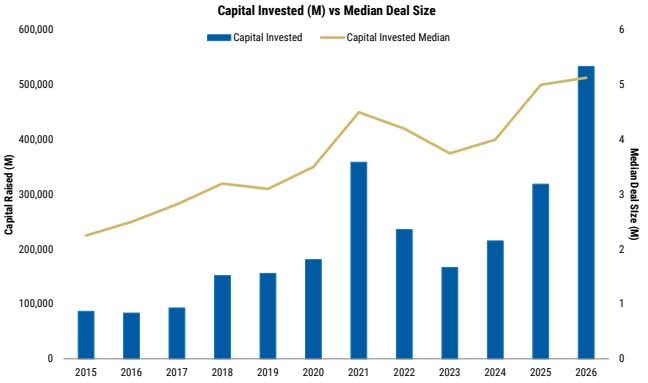
Exhibit 13: Re-Imagine the Data Center

Company	Sub-Layer	Valuation ($m)	Valuation date	Lead investors (latest priced round)	Company description
Crusoe	Energy-Integrated Compute Platforms	$30,900	Sep 3, 2026	Valor Equity Partners, Atreides, Mubadala Capital (Series F, Sep 2026)	Operator of artificial intelligence-optimized cloud infrastructure intended to deliver computing for next-generation artificial intelligence workloads.
Armada	Distributed & Modular Data Centers	$2,340	May 19, 2026	8090 Industries, Overmatch Ventures, OVTR VC, BlackRock (Series B, May 2026)	Developer of an edge computing platform designed to revolutionize connectivity, computing, and artificial intelligence tools.
Starcloud	Space-Based Data Centers	$2,300	Aug 21, 2026	EQT, The Merry Ventures, Manhattan West, Benchmark Opportunity Partners (Series A, Aug 2026)	Developer of power and cooling systems designed to create cost-competitive, sustainable, and rapidly scalable data centers in space.
Cowboy Space	Space-Based Data Centers	$2,029	May 11, 2026	Index Ventures (Series B, May 2026)	Developer of orbital computing infrastructure designed to provide space-based power generation and artificial intelligence compute capacity.
Rune	Distributed & Modular Data Centers	$142	Sep 16, 2026	Spark Capital (Series A, Sep 2026)	Developer of modular micro data centers designed to support compute infrastructure at renewable energy sites.
Nautilus Data Technologies	Environment-Coupled Facilities	$111	Jan 27, 2026	Pangaea Ventures (Series B, Jan 2026)	Developer of a water-cooled data center designed to set a standard for energy efficiency, environmental sustainability, and global scalability.
AVNOS	Environment-Coupled Facilities	$110	Dec 4, 2023	NextEra Energy Resources (Series A, Dec 2023)	Developer of carbon capture and water harvesting technology designed for climate mitigation purposes.
Flexnode	Distributed & Modular Data Centers	$93	Jan 29, 2026	No lead named: Eaton, Colle Capital Partners, Activant Capital (Series A, Jan 2026)	Developer of modular data centers designed to support high-density computing.
Lonestar Data Holdings	Space-Based Data Centers	$90	Oct 31, 2025	No lead named: Gaingels, Abbey Road Investment Group (Series A, Oct 2025)	Developer of space-based data storage infrastructure designed to ensure secure and resilient data continuity beyond terrestrial systems.
Sophia Space	Space-Based Data Centers	$40	Dec 31, 2025	EverGreen (NVIDIA Alumni Investment Network) (SAFE, Jun 2026)	Operator of a solar-powered data processing platform intended to alleviate the constraints faced by terrestrial data centers.
Northstar	Distributed & Modular Data Centers	$31	Jun 10, 2022	Undisclosed investors (Series A, May 2023)	Developer of modular structural systems designed to address complex infrastructure and deployment challenges across multiple environments.
ECL (Systems and Information Management)	Distributed & Modular Data Centers	$28	Feb 10, 2022	No lead named: Hyperwise Ventures, Bachmanity Capital (2024)	Operator of sustainable data centers intended for cloud and deployments, and in the delivery of off-grid, reliable, sustainable, cost-effective hydrogen-powered data centers-as-a-service.
Liquidstar	Distributed & Modular Data Centers	$18	Aug 18, 2022	No lead named: Orange DAO (Jan 2025)	Developer of a decentralized infrastructure system designed to deliver energy, water, connectivity, and computing capabilities in off-grid environments.
GridFree AI	Distributed & Modular Data Centers	$15	Feb 4, 2025	Giant Ventures (Seed, Feb 2025)	Developer of a computing infrastructure platform intended to offer novel end-to-end data center facilities.

Source: Pitchbook, Morgan Stanley Research. Up to 15 companies by latest known valuation (post-money), associated with completed rounds.

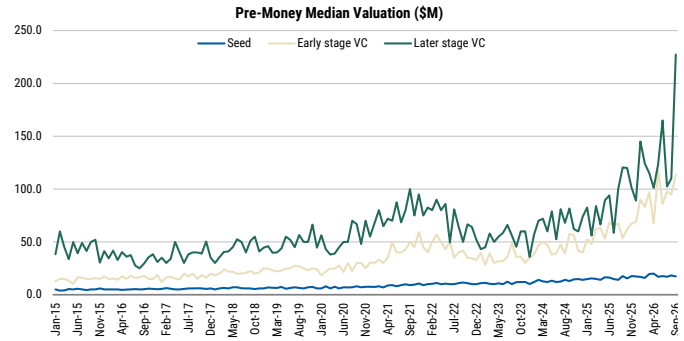
State of Venture in Charts

Exhibit 14: Annual (YTD) Capital Raised vs Median Deal size



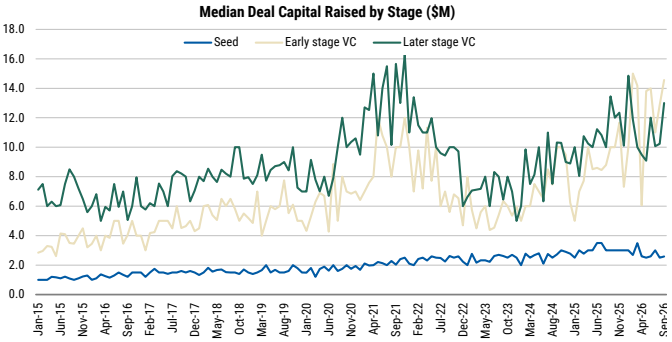
Source: PitchBook as of 10/6/2026. Annual or YTD.

Exhibit 15: Median Pre-Money Valuation By Stage



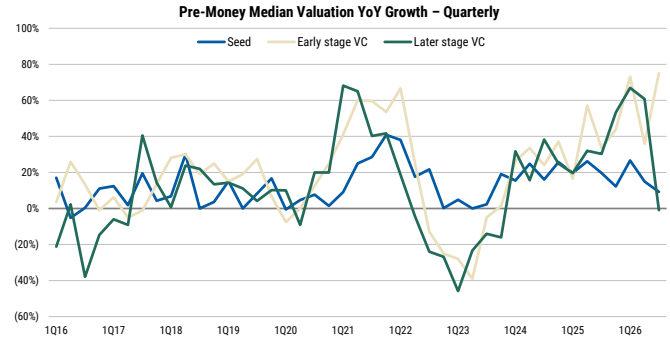
Source: PitchBook as of 10/6/2026. Monthly.

Exhibit 16: Median Deal Capital Raised by Stage



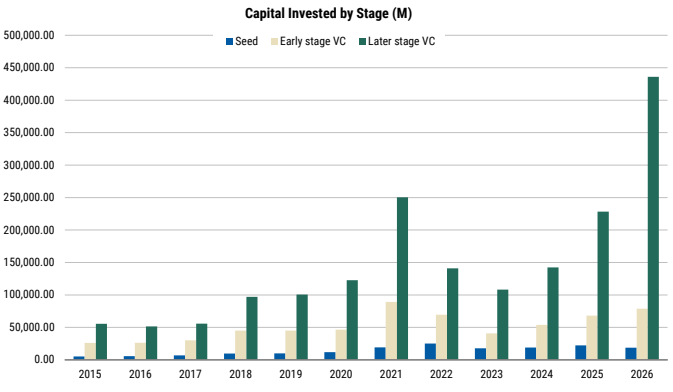
Source: PitchBook as of 10/6/2026. Monthly.

Exhibit 17: Pre-Money Median Valuation YoY Growth by Stage



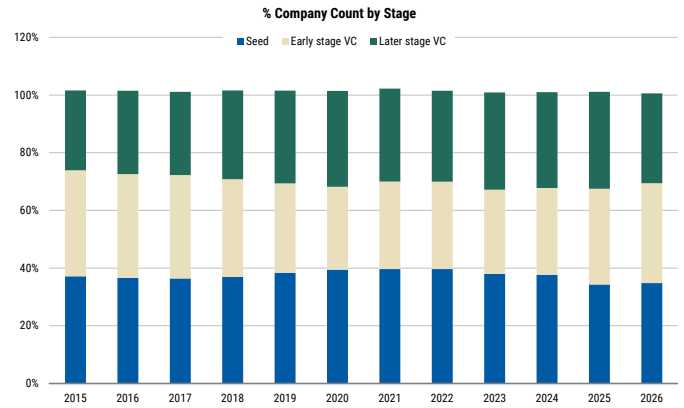
Source: PitchBook as of 10/6/2026. Quarterly.

Exhibit 18: Capital Invested by Stage



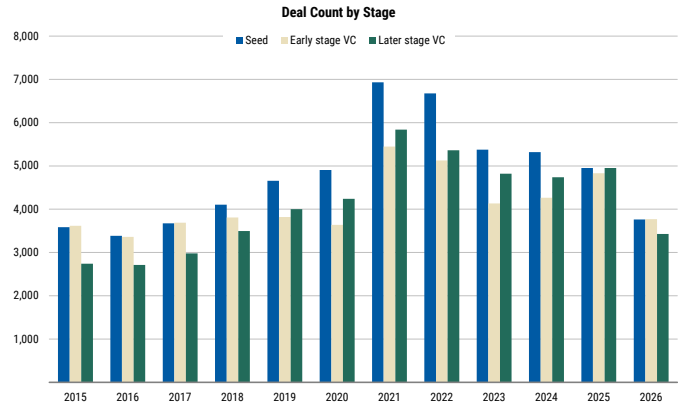
Source: PitchBook as of 10/6/2026. Annual or YTD.

Exhibit 19: Capital Invested by Stage %



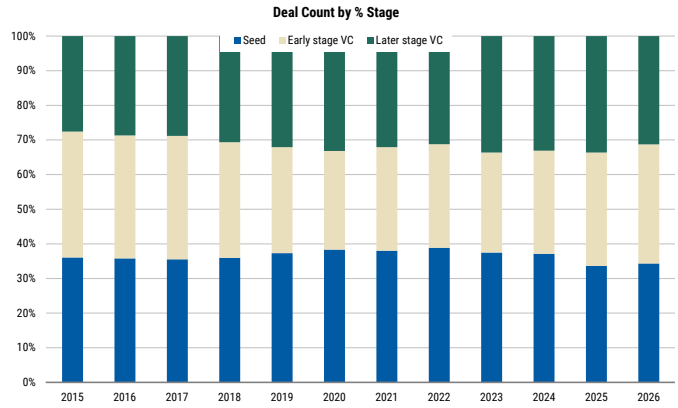
Source: PitchBook as of 10/6/2026. Annual or YTD.

Exhibit 20: Deal Count by Stage



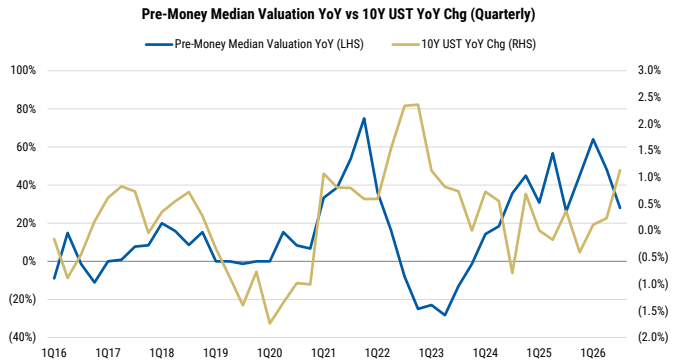
Source: PitchBook as of 10/6/2026. Annual or YTD.

Exhibit 21: Deal Count by % Stage



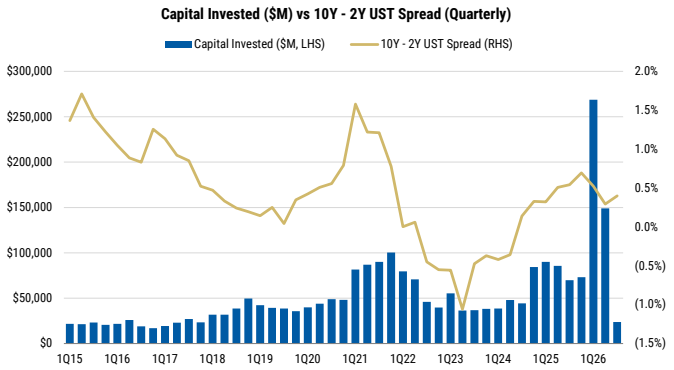
Source: PitchBook as of 10/6/2026. Annual or YTD.

Exhibit 22: Median Pre-Money Valuation % Change YoY vs 10Y UST YoY



Source: PitchBook as of 10/6/2026. Quarterly

Exhibit 23: Capital Raised vs 2s10s UST Curve

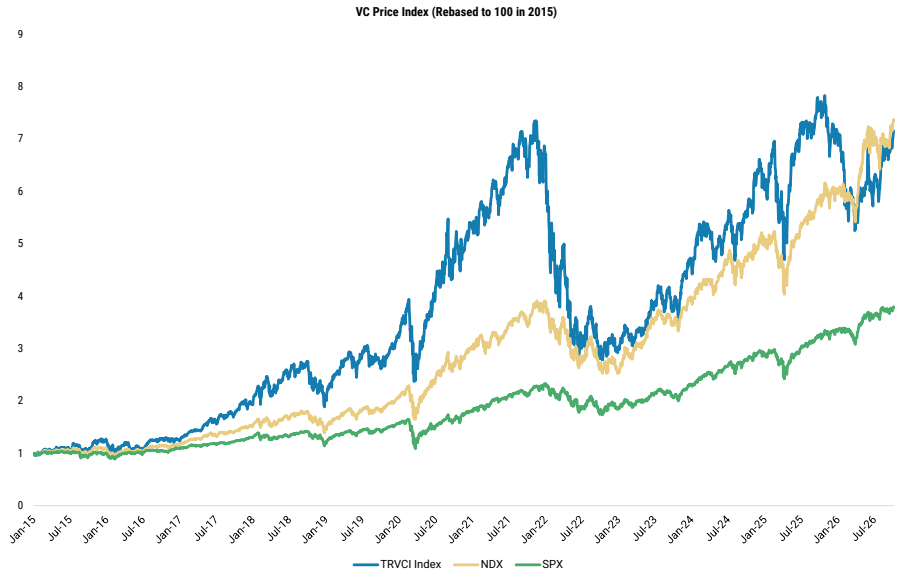


Source: PitchBook as of 10/6/2026. Quarterly

TRVCI Index

- Top-down performance proxy for VC-backed companies by LSEG
- Sector weights are large drivers of index performance, as of Sep 2026 the index has 76% technology - software exposure.

Exhibit 24:



Source: Bloomberg as of 10/7/2026, Morgan Stanley Research

US Sector Private Company Research

Our sector and thematic teams share interviews and provide in-depth research on private companies

Aerospace and Defense

- Defense: Dollars, Disruptors, and De(Consolidation); Identifying Investment Opportunities (6 Mar 2026)
- Defense Tech: Anduril: a NeoPrime(r) (12 Jan 2026)

Autos & Shared Mobility

- Industrial Edge: Enabling Industrial Autonomy: Applied Intuition (28 Apr 2026)

Healthcare

- Transforming Healthcare Private Company Research Series: Latest Checks On Ambient AI Scribe and LLM Use In Health Systems (20 Jul 2026)
- Transforming Healthcare: Private Company Research Series: Discussions with the Founders of Commure and Ambience (6 Apr 2026)
- Transforming Healthcare: HIMSS Global Healthcare Conference: An inflection point for AI adoption (16 Mar 2026)
- Transforming Healthcare: Private Company Research Series: A Deeper Dive into OpenEvidence (12 Jan 2026)

Metals & Mining

- Rare Earths: Where Is The Money Flowing?

Pharmaceuticals & Biotechnology

- Biotech Before the Bell: Takeaways from Morgan Stanley's Private Company Conference (26 Jan 2026)

Robotics, Humanoids & Physical AI

- Robotics: RoboFest California Preview: Bots by the Bay (16 Sep 2026)
- Robotics: Actuators: Robot Muscle Needs American Hustle (13 Sep 2026)
- Robotics: When Factory = Robot (26 Mar 2026)
- Embodied AI: World Models: AI's Journey from Digital to Physical (22 Mar 2026)
- Robotics: Agile Robots: Das Bot (17 Mar 2026)
- Rowdy Robot: RoboFest Silicon Valley Takeaways, 'AlphaGo' Moment for AVs (23 Jan 2026)
- Robotics: Bedrock Robotics: The Bots Break Ground (12 Jan 2026)

Software

- Databricks: First Data, Today AI, Next the Agentic Enterprise (18 Sep 2026)
- Video: Inside the Office of the CFO: Puzzle CEO & Co-Founder, Sasha Orloff (30 Apr 2026)

- Software: Canva Create 2026: Building the Creative Operating System (21 Apr 2026)
- Software: Pendo.io: Building a Platform for Product Experience and Agent Analytics (31 Mar 2026)

Space

- Space: Varda Space: Space Manufacturing, Pharma Made in Orbit (19 Feb 2026)
- Space: Starcloud: Data Centers in Space - AI Goes Orbital (27 Jan 2026)

Sports

- Sports: The Fanatics Playbook (Primer): Building the Business of Fandom (22 Sep 2026)

Sustainability

- Sustainability: Earthshots Series: Enhancing EV Efficiency with Advanced Tyres (7 May 2026)
- Sustainability: Earthshots Series: Sodium-Ion Battery Technology (16 Apr 2026)

VC Glossary

- **Pre-money valuation**: The value of a company before a new financing round (pre-money valuation = post-money valuation – round amount).
- **Post-money valuation:** The value of a company after new capital is invested (pre-money valuation + new investment).
- **Down round:** A financing round completed at a lower valuation than the previous round.
- **Seed Stage** (PitchBook): When any type of investor provides the initial financing for a new enterprise that is in the earliest stages of development. PitchBook only designates an equity raise as seed financing when it is explicitly referenced as a seed deal in sources or when undisclosed investors invest in a company under a set of predetermined circumstances.
- **Early Stage** (PitchBook): An Early Stage VC deal is defined as a series A to series B round that occurs within five years of the company's founding date. We also categorize the deal as Early Stage VC if no series is associated with the deal and the deal happens within five years of the company's founding date.
- **Late Stage** (PitchBook): This deal type includes Later Stage VC and Venture Growth deals. A Later Stage VC deal is defined as a series C to series D round, or a round that occurs more than five years after the company's founding date. A Venture Growth deal is defined as a series E+ round, or a round founded more than seven years ago with six or more VC deals.
- **Unicorn**: A private startup valued at $1 billion or more.
- **Decacorn**: A private company valued at $10 billion or more.
- **Dry powder:** Capital that VC funds have raised but not yet invested.
- **Exit**: A liquidity event, such as an IPO, acquisition, merger, or secondary sale.
- **Strategic Acquisition**: An acquisition by an operating company primarily to enhance its own core business—such as expanding markets, gaining technology, or achieving synergies—rather than just seeking a financial return.
- **Financial Acquisition (or buyout)**: An acquisition by an investment-focused entity (e.g., PE firm, asset manager, family office) primarily to generate returns for investors through improving the company and exiting via sale, IPO, or recapitalization.
- **Secondary Sale:** A transaction in which an existing investor sells their ownership interest to another buyer, rather than the company issuing new shares. In private equity or venture capital, this may involve selling shares in a private company or selling an interest in a fund. The proceeds go to the selling investor, not usually to the company.

Stocks mentioned — Astera Labs (ALAB.O, current price $382.25) and Cerebras Systems (CBRS.O, $175.05), both covered by Joe Moore; Bloom Energy (BE.N, $291.29), covered by Dave Arcaro; SpaceX (SPCX.O, $167.60), covered by Adam Jonas.

Valuation Methodology and Risks

Astera Labs Inc (ALAB.O)

We apply a 0.37x CY27 EV/sales/growth multiple, assuming a 62% revenue CAGR. While this represents a premium to AI peers, we believe it is warranted by ALAB's superior growth profile and nearly 100% exposure to AI connectivity.

Risks to Upside

- Increase AI spend to propel data center revenue
- Quicker upgrade of underlying technology driving content growth
- New product launches increasing TAM

Risks to Downside

- Increased competition reduces Astera's market share meaningfully
- Pause in AI spend and data center investment halts future revenue growth
- CXL servers and 1.6T port speeds are significantly delayed, preventing new products from gaining traction

Bloom Energy Corp. (BE.N)

Derived from an unlevered DCF analysis with 10.8% discount rate, based on a 4.0% risk-free rate, 1.75 Beta and 4.29% equity risk premium.

Risks to Upside

- Higher revenue growth through '35, given current low market penetration rates
- Cost reduction initiatives outpace expectations
- Customer adoption increase in light of de-energization in CA.

Risks to Downside

- Regulators impose charges on C&I customers that use distributed generation.
- Stricter emissions regulation.
- New entrants / technologies emerge with cost-competitive products.
- Cost reductions don't materialize.

Cerebras Systems (CBRS.O)

13x multiple on our 2028 adjusted revenue of $6.3bn. We comp Cerebras to AI semis cohort and 13x is in-line with with smid cap AI names with near 100% AI exposure that are outgrowing the overall market.

Risks to Upside

- Contract Extension
- Additional customer
- Capacity deployment

Risks to Downside

- Capacity deployment/infrastructure risk

- Customer concentration risk
- Competitive risk

SpaceX (SPCX.O)

Sum-of-the-Parts by Segment: Space, Connectivity, AI (Divided into X & Grok and Enterprise AI)

- $300 Price Target = sum of Space ($8), Connectivity ($118), X & Grok ($8), Enterprise AI ($165)

2040 ending forecast period

6/30/2027 valuation date

11.2% WACC; 11.9% Cost of Equity.

50% Enterprise AI valuation discount for execution risk

TGR of: 4.0% Space, 4.5% connectivity, 3.0% X & Grok, 5.0% Enterprise AI

Equates to 0.46 EV/EBIT/Growth

Risks to Upside

- Faster Starship reuse progress
- Faster Starlink capacity growth
- Stronger DTC / enterprise adoption
- More neocloud wins
- Cursor ARR acceleration
- Lower AI infrastructure time to power and cost

Risks to Downside

- Slower Starship reuse cadence
- Slower Starlink subscriber growth
- Weaker enterprise AI monetization
- Higher capex / cost per watt of compute
- Longer time-to-power
- Greater funding needs / dilution
- Regulatory delays

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Global Stock Ratings Distribution

(as of September 30, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

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Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

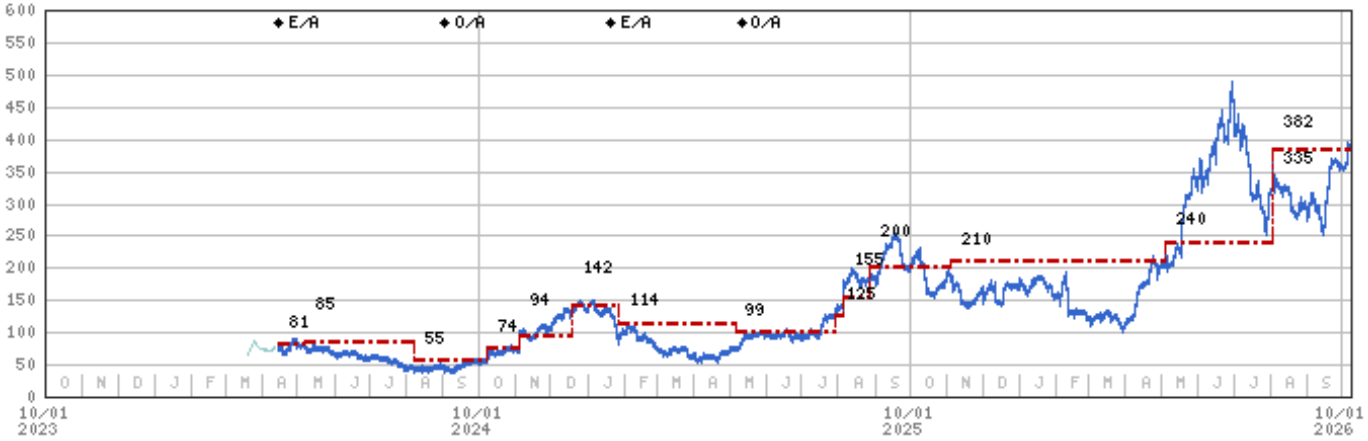
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Astera Labs Inc (ALAB.O) - As of 10/08/26 GMT in USD
Industry : Semiconductors



Stock Rating History: 4/15/24 : E/A; 9/3/24 : O/A; 1/20/25 : E/A; 5/11/25 : O/A

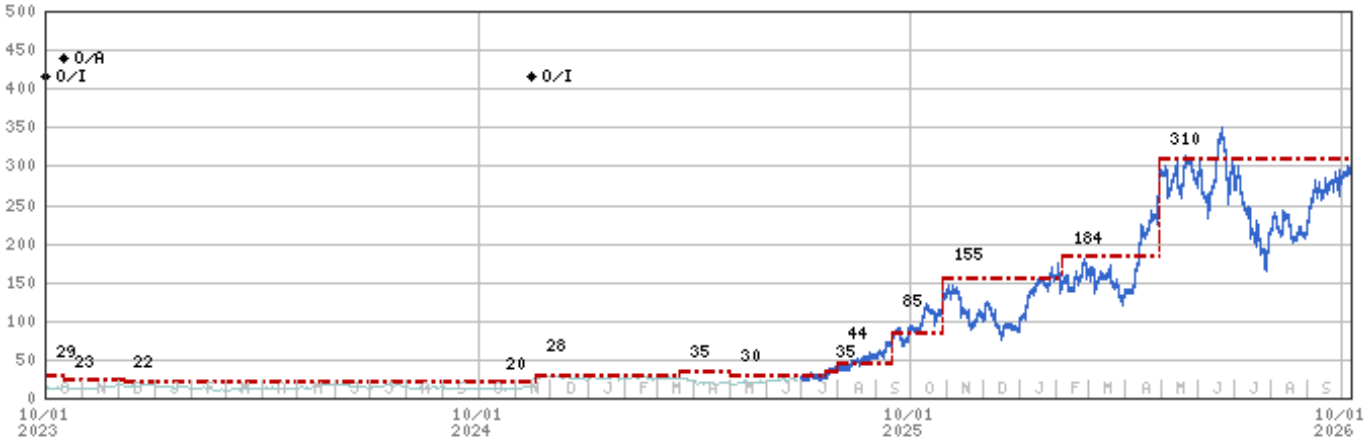
Price Target History: 4/15/24 : 81; 5/7/24 : 85; 8/7/24 : 55; 10/8/24 : 74; 11/4/24 : 94; 12/19/24 : 142; 1/27/25 : 114; 5/6/25 : 99; 7/29/25 : 125; 8/6/25 : 155; 8/28/25 : 200; 11/4/25 : 210; 5/5/26 : 240; 8/3/26 : 335; 8/4/26 : 382

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -·- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) ▬
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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Bloom Energy Corp. (BE.N) - As of 10/08/26 GMT in USD
Industry : Clean Tech



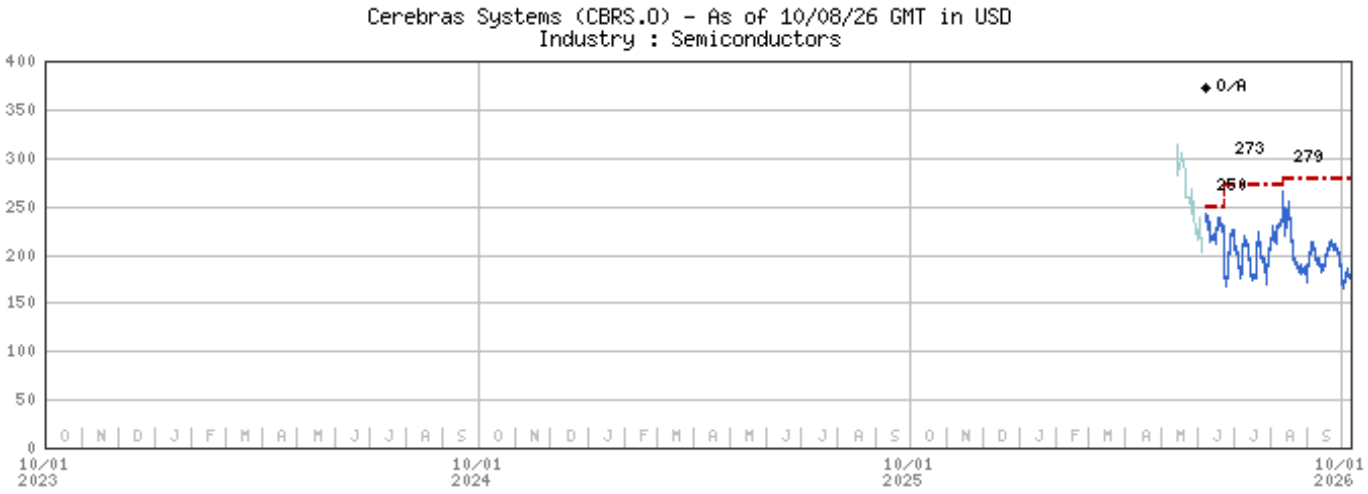
Stock Rating History: 10/1/21 : E/I; 1/13/22 : E/A; 1/9/23 : O/A; 4/3/23 : O/I; 10/17/23 : O/A; 11/15/24 : O/I

Price Target History: 8/19/21 : 25; 10/26/21 : 32; 1/13/22 : 26; 4/4/22 : 25; 6/13/22 : 21; 8/18/22 : 37; 1/9/23 : 35; 4/3/23 : 32; 5/16/23 : 30; 8/31/23 : 29; 10/17/23 : 23; 12/6/23 : 22; 10/15/24 : 20; 11/18/24 : 28; 3/20/25 : 35; 5/1/25 : 30; 7/21/25 : 35; 8/1/25 : 44; 9/16/25 : 85; 10/29/25 : 155; 2/6/26 : 184; 4/29/26 : 310

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -·- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) ▬
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
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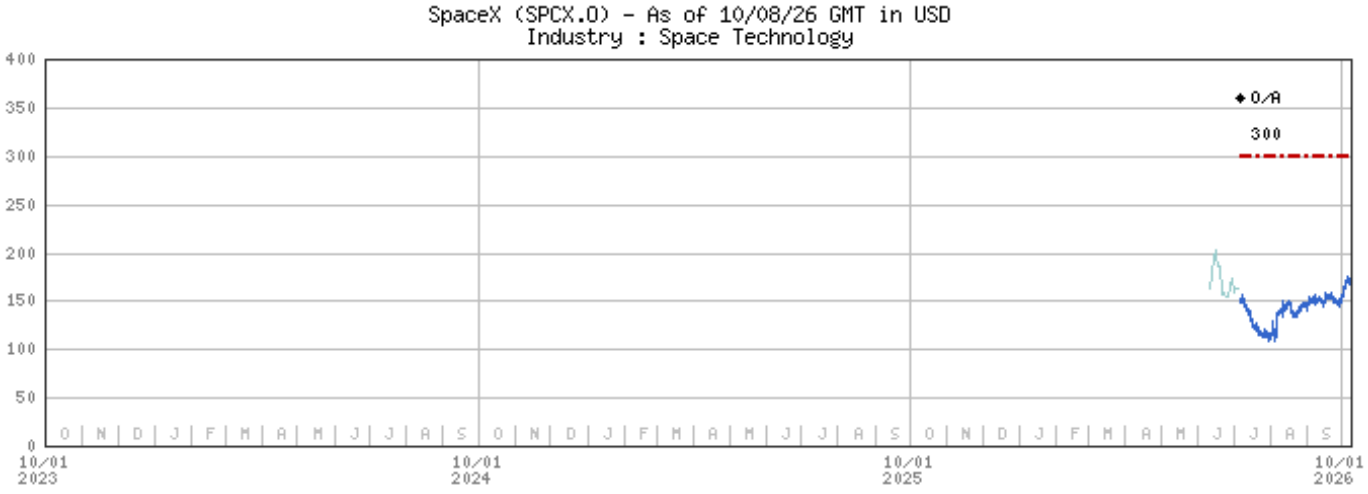
Stock Rating History: 6/8/26 : O/A

Price Target History: 6/8/26 : 250; 6/23/26 : 273; 8/12/26 : 279

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -·- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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Stock Rating History: 7/7/26 : O/A

Price Target History: 7/7/26 : 300

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -·- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

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